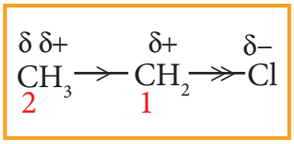
basicity etc., are affected by the displacement of electrons that takes place in its covalent bonds. This movement can be influenced by either the atoms/groups present in close proximity to the bond or when a reagent approaches a molecule. The displacement effects can either be permanent or a temporary. In certain cases, the electron displacement due to an atom or a substituent group present in the molecule cause a permanent polarisation of the bond and it leads to fission of the bond under suitable conditions. The electron displacements are catagorised into inductive effect (I), resonance effect (R), electromeric effect (E) and hyper conjugation.

Inductive effect (I)

Inductive effect is defined as the change in the polarisation of a covalent bond due to the presence of adjacent bonds, atoms or groups in the molecule. This is a permanent phenomenon.

Let us explain the inductive effect by considering ethane and ethylchloride as examples. The C-C bond in ethane is non polar while the C-C bond in ethyl chloride is polar. We know that chlorine is more electronegative than carbon, and hence it attracts the shared pair of electron between C-Cl in ethyl chloride towards itself. This develops a slight negative charge on chlorine and a slight positive charge on carbon to which chlorine is attached. To compensate it, the C_1 draws the shared pair of electron between itself and C_2. This polarisation effect is called inductive effect. This effect is greatest for the adjacent bonds, but they also be felt farther away. However, the magnitude of the charge separation decreases rapidly, as we move away from C_1 and is observed maximum for 2 carbons and almost insignificant after 4 bonds from the active group.



It is important to note that the inductive effect does not transfer electrons from one atom to another but the displacement effect is permanent. The inductive effect represents the ability of a particular atom or a group to either withdraw or donate electron density to the attached carbon. Based on this ability the substituents are classified as +I groups and -I groups. Their ability to release or withdraw the electron through sigma covalent bond is called +I effect and -I effect respectively.

Highly electronegative atoms and groups with an atom carrying a positive charge are electron withdrawing groups or –I groups.

Example: -F, -Cl, -COOH, -NO_2, -NH_2

Higher the electronegativity of the substitutent, greater is the -I effect. The order of the –I effect of some groups are given below.

NH_3> NO_2> CN > SO_3H > CHO > CO > COOH > COCl > $CONH_2$> F > Cl > Br > I > OH > OR > NH_2> C_6H_5> H

Highly electropositive atoms and atoms are groups which carry a negative charge are electron donating or +I groups.

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